
Finding the Informative Frontier: Example Weighting for LoRA Fine-Tuning

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Abstract

1 Fine-tuning pre-trained models on downstream tasks treats all training exam-
2 ples equally, yet their informativeness changes continuously as the model adapts.
3 Samples the model has already mastered contribute near-zero gradients; samples
4 whose labels conflict with the model’s strong pre-trained prior never converge
5 and contribute only noise. We propose **EWL (Example Weighting via Learning
6 Progress)**, which identifies and upweights the *adaptation frontier*: examples where
7 loss is actively declining and gradient signal is richest. EWL tracks per-sample
8 loss velocity via exponential moving averages and multiplies by a *LoRA gradient
9 proxy* that acts as an adaptive temperature: large during rapid adaptation, decay-
10 ing near convergence, providing a free automatic curriculum that approximates
11 optimal importance weights for gradient estimation. It requires no meta-gradients,
12 no held-out data, and adds no inference cost. Across three fine-grained visual
13 benchmarks (FGVC-Aircraft, CUB-200, Food-101) and four instruction-tuning
14 datasets (Alpaca, Dolly-15k, OASST1, FLAN-Mini) with Llama at 1B and 8B
15 scale, EWL consistently improves over standard LoRA SFT by **1.4–2.8%** accuracy
16 and **1.5–2.9%** ROUGE-L with a single interpretable hyperparameter.

17 1 Introduction

18 Fine-tuning a pre-trained transformer is not like training from scratch. The model arrives with
19 powerful, general-purpose representations forged from vast data, and the fine-tuning corpus is small
20 by comparison. This creates a fundamental asymmetry that uniform gradient averaging ignores: the
21 *informativeness* of a training sample is not fixed; it changes continuously as the model adapts.

22 Consider what happens to a training sample’s gradient over time. Early in fine-tuning, many samples
23 are genuinely novel to the model; their gradients are large and directionally consistent across steps.
24 As training progresses, samples inside the model’s existing decision boundaries become redundant:
25 their loss has already dropped, their gradients are near-zero, and they dilute the mini-batch gradient
26 with noise. Meanwhile, samples whose labels are inconsistent with the model’s pre-trained prior
27 stagnate at high loss. Their gradients point in directions the model cannot reconcile, contributing
28 variance without reducing loss. The only samples adding real information at any given step are those
29 at the **adaptation frontier**: examples where loss is actively declining, indicating that the model is
30 successfully reconfiguring its representations.

31 This observation has strong empirical grounding. Arpit et al. [1] showed that deep networks prioritise
32 learning generalising patterns before memorising noise; loss trajectories differ predictably between
33 learnable and unlearnable examples. Toneva et al. [2] demonstrated that most training examples
34 are “unforgettable,” learned early and never revisited, while a small, near-boundary set is repeatedly
35 contested. Most compellingly, Swayamdipta et al. [3] operationalised this as *Dataset Cartography*:
36 training examples can be mapped by their mean confidence and variability across epochs into easy,

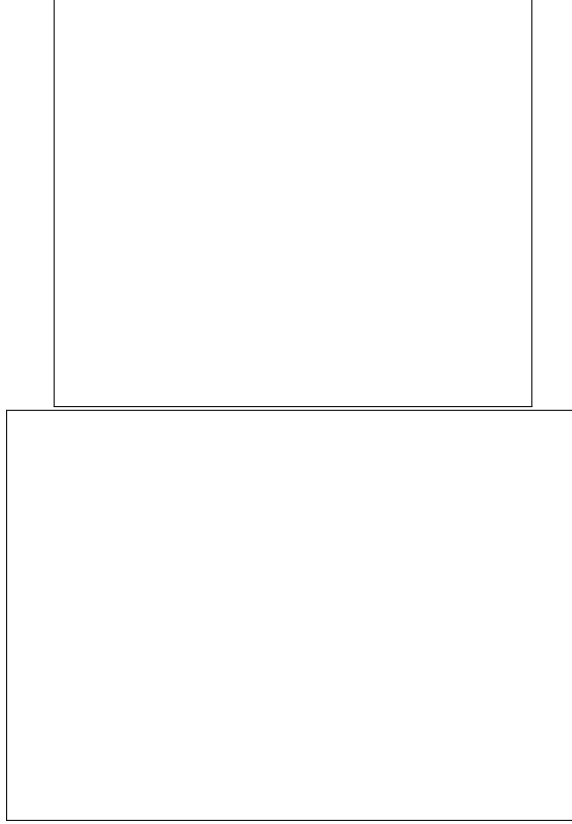


Figure 1: **Left:** Loss trajectories for three representative sample types. The EWL progress signal $(\mu_i - \ell_i)/\mu_i$ is large and positive *only* for samples whose loss is actively declining (the adaptation frontier). Samples already mastered and samples conflicting with the pre-trained prior both produce near-zero signals, suppressing their gradient contribution automatically and without supervision. **Right (illustration):** Predicted distribution of training samples in mean-EMA $\times$ loss-variability space, analogous to Dataset Cartography Swayamdipta et al. [3]. EWL’s time-averaged weights are predicted to concentrate on the high-variability cluster, the stratum that Swayamdipta et al. [3] showed outperforms the full dataset when used alone. Empirical validation via Prediction 2 (§5).

37 ambiguous, and hard-to-learn strata. They found that training on only the *ambiguous* stratum (high
 38 variability in model predictions, $\approx 30\%$ of data) *outperforms training on the full dataset*. The
 39 adaptation frontier is not just a useful heuristic; it is where the data’s value concentrates.

40 **Our contribution.** EWL recovers the benefit of selecting the ambiguous stratum *online*, in a single
 41 training pass, without a preliminary epoch, without held-out data, and without bi-level optimisation.
 42 The method is grounded in two complementary principles:

43 **(i) Gradient variance reduction.** Weighting samples by their loss rate of change is an approximation
 44 to optimal importance sampling for mini-batch gradient estimation. Samples at the adaptation frontier
 45 provide low-variance, directionally consistent gradients; stagnant samples contribute high variance
 46 with no expected reduction in loss.

47 **(ii) LoRA subspace alignment.** LoRA constrains updates to a rank- r subspace. EWL’s progress
 48 signal naturally selects for examples whose gradients have non-trivial projections *within* this subspace,
 49 the only gradients the adapter can act on. This makes EWL particularly well-matched to parameter-
 50 efficient fine-tuning in a way that loss-magnitude heuristics are not.

2 Related Work

Training dynamics and Dataset Cartography. Swayamdipta et al. [3] characterize NLP fine-tuning examples by confidence and variability across epochs, finding three strata (easy, ambiguous, hard-to-learn) and showing the ambiguous stratum is most valuable for generalisation. Arpit et al. [1] establish mechanistically that networks learn generalising patterns before memorizing noise. Toneva et al. [2] show that “forgettable” examples near decision boundaries drive generalization, while unforgettable examples plateau early. EWL directly operationalizes these findings in a single online pass, without pre-computing corpus statistics or requiring a clean validation set.

Curriculum and self-paced learning. Bengio et al. [4] showed easy-to-hard ordering accelerates learning; Kumar et al. [5] made this adaptive via a self-paced regularizer. Both weight by loss *magnitude*: they persistently up-weight high-loss samples, conflating informative frontier examples with irreconcilable noise. EWL weights by loss *rate of change*, which resolves this ambiguity.

Noise robustness and hard example mining. Jiang et al. [6] and Han et al. [7] address noisy labels via a mentor network or mutual small-loss selection, both requiring a clean reference set or doubled compute. Shrivastava et al. [8] and Lin et al. [9] mine hard examples by loss magnitude, up-weighting high-loss samples regardless of whether they are genuinely informative or simply irreconcilable noise. EWL achieves implicit noise suppression without any explicit noise model: a sample whose loss never declines produces a near-zero progress signal and is automatically down-weighted. The key distinction is that EWL uses loss *velocity*, which is orthogonal to loss magnitude and can separate frontier examples from noise where magnitude-based methods cannot.

Meta-learning for reweighting. Ren et al. [10] and Shu et al. [11] learn optimal per-example weights via meta-gradient descent on a held-out clean set. Both are highly effective but impose second-order gradient computation (roughly doubling wall-clock time) and require a trusted validation split. EWL requires neither.

LoRA and parameter-efficient fine-tuning. LoRA [12] injects trainable low-rank perturbations $\Delta W = BA$ into frozen pre-trained weights. AdaLoRA [13] extends this by using the Frobenius norm of adapter *weight* matrices as importance scores to dynamically reallocate rank budget across layers. EWL uses Frobenius norms of adapter *gradients* for a different purpose: as a step-level proxy for adapter activity that modulates weighting sharpness over time rather than structural rank. The interaction between EWL and LoRA’s low-rank constraint is a novel design point discussed in section 3.4.

3 Method

3.1 Progress Signal

Let $\ell_i^{(t)}$ be the per-sample cross-entropy at step t . EWL maintains a per-sample EMA $\mu_i^{(t)}$ as a smoothed loss history, serving as a cheap online surrogate for the per-epoch confidence statistics used by Dataset Cartography [3]. The **relative progress signal** is:

$$\mu_i^{(t)} = \alpha \mu_i^{(t-1)} + (1-\alpha) \ell_i^{(t)}, \quad (1)$$

$$s_i = \underbrace{\frac{\mu_i^{(t-1)} - \ell_i^{(t)}}{|\mu_i^{(t-1)}| + \varepsilon}}_{\text{velocity}} \cdot \mu_i^{(t)}, \quad \varepsilon = 10^{-8}. \quad (2)$$

The numerator is positive when loss is below its smoothed history, indicating the model is making progress on this sample. Dividing by μ makes the signal **scale-invariant**: a drop from 3.5 to 3.0 and from 0.35 to 0.30 both yield $s^{\text{rel}} \approx 0.14$, allowing EWL to run unchanged across vision (loss $\in [0.01, 5]$) and language (loss $\in [1, 3]$) tasks.

The suppression of non-frontier samples follows directly from the formula. Mastered samples have $\mu \approx \ell \approx 0$, so $s^{\text{rel}} \approx 0$. Samples conflicting with the model’s prior have $\ell \approx \mu \gg 0$, so again $s^{\text{rel}} \approx 0$ or slightly negative. Only the adaptation frontier, where $\ell < \mu$, produces a consistently positive

Algorithm 1 EWL Training Step

Require: Mini-batch $\{(x_i, y_i)\}$, EMA state $\{\mu_i\}$, lagged proxy g_{prev} , temperature τ , warmup K

- 1: Compute per-sample losses $\{\ell_i\}$
- 2: **if** step $\leq K$ **then return** uniform weights $1/B$
- 3: **end if**
- 4: $\mu_i^{\text{prev}} \leftarrow \mu_i$ $\triangleright$ snapshot before update
- 5: $\mu_i \leftarrow \alpha \mu_i + (1-\alpha) \ell_i$ $\triangleright$ EMA update
- 6: $s_i \leftarrow \underbrace{\frac{\mu_i^{\text{prev}} - \ell_i}{\mu_i^{\text{prev}} + \varepsilon}}_{\text{velocity}} \cdot \underbrace{\mu_i}_{\text{magnitude}} \cdot \underbrace{g_{\text{prev}}}_{\text{adaptive temp.}}$
- 7: $w_i \leftarrow \text{softmax}(s/\tau)$
- 8: Backprop $\mathcal{L} = \sum_i w_i \ell_i$
- 9: $g_{\text{prev}} \leftarrow \frac{1}{|\Lambda|} \sum_{\theta \in \Lambda} \|\nabla_{\theta}\|_F$ $\triangleright$ mean Frobenius norm over all LoRA param tensors Λ

94 signal. This distinguishes EWL from self-paced [5] and focal loss [9] methods, which weight by loss
 95 *magnitude* and therefore conflate the frontier with irreconcilable noise.

96 3.2 LoRA Gradient Proxy

97 To capture the current magnitude of adapter activity and gate the progress signal by how much the
 98 model is updating at this step, we multiply by a **LoRA gradient proxy**:

$$g^{(t)} = \frac{1}{|\Lambda|} \sum_{\theta \in \Lambda} \|\nabla_{\theta}\|_F, \quad (3)$$

99 where $\mathcal{A}$ is the set of LoRA adapter pairs $(B, A) \in \mathbb{R}^{d \times r} \times \mathbb{R}^{r \times k}$ [12] across all layers and $\|\cdot\|_F$
 100 denotes the Frobenius norm. This is the gradient analogue of the weight-magnitude importance
 101 score used by AdaLoRA [13] for rank allocation; here it is applied to modulate weighting sharpness
 102 rather than structural rank. The combined signal is $s_i = s_i^{\text{rel}} \times g^{(t-1)}$, where we use the *lagged*
 103 proxy (computed after the previous backward pass) to avoid circular dependency. When adapters are
 104 updating rapidly (large g), the signal is amplified and weights become sharper; when adapters have
 105 converged ($g \rightarrow 0$), weights collapse to uniform, gracefully recovering the SFT baseline.

106 3.3 Weight Computation

107 The signal s_i is passed directly to a softmax with temperature τ , following the importance-sampling
 108 framework of Katharopoulos and Fleuret [14]:

$$w_i = \frac{e^{s_i/\tau}}{\sum_j e^{s_j/\tau}}. \quad (4)$$

109 The weighted loss is $\mathcal{L}_{\text{EWL}} = B \sum_i w_i \ell_i$. At $\tau \rightarrow \infty$ weights are uniform (EWL $\equiv$ SFT); at $\tau \rightarrow 0$
 110 all mass concentrates on one sample. Importantly, *z-score normalisation is intentionally omitted*. For
 111 any positive scalar g , $(g \cdot x - g\bar{x})/\sigma_{gx} = (x - \bar{x})/\sigma_x$, so normalising before the softmax would cancel
 112 the grad-proxy modulation entirely. The grad proxy therefore acts as an **adaptive temperature**: it
 113 implicitly scales $1/\tau_{\text{eff}} = g^{(t-1)}/\tau$, producing sharper weights during rapid early adaptation and
 114 flatter weights near convergence. This provides a free automatic curriculum with no additional
 115 hyperparameter. A K -step warmup ($w_i = 1/B$) lets the EMA accumulate meaningful history.

116 3.4 Theoretical Grounding

117 **Gradient variance reduction.** Mini-batch gradient estimation approximates the population gradient
 118 $\mathbf{g}^* = \mathbb{E}[\nabla_{\theta} \ell]$ from a noisy sample. The optimal importance weights for minimising estimator variance
 119 are proportional to $\|\nabla_{\theta} \ell_i\|$ [14], the gradient magnitude of each sample. EWL’s progress signal
 120 $(\mu_i - \ell_i)/\mu_i$ is a low-cost proxy for this: when a sample’s loss is actively declining, its gradient is
 121 large and directionally consistent with the current descent direction. Letting $\mathbf{g}_i = \nabla_{\theta} \ell_i$, the EWL

Table 1: Top-1 accuracy on three fine-grained visual benchmarks. EWL improves consistently; FGVC-Aircraft shows the largest gain because 100 aircraft variants share nearly identical silhouettes. The fraction of examples at the adaptation frontier at any moment is large, as many samples lie near rapidly-shifting decision boundaries. This prediction from the gradient variance reduction perspective is borne out by the data.

Dataset	SFT Baseline	EWL	Δ	τ^*
FGVC-Aircraft	71.7%	74.5%	+2.8%	2.0
CUB-200-2011	86.9%	88.6%	+1.7%	2.0
Food-101	87.1%	88.5%	+1.4%	1.5

122 update

$$\Delta\theta \propto \sum_i w_i \mathbf{g}_i \quad \text{vs.} \quad \Delta\theta \propto \frac{1}{B} \sum_i \mathbf{g}_i \quad (5)$$

123 concentrates mass on $\mathbf{g}_i$ with high expected magnitude, reducing the variance of the gradient estimator
 124 without requiring explicit gradient magnitude computation. This argument holds regardless of the
 125 choice of optimiser (Adam, AdamW, etc.) since the variance is a property of the data-sampling step,
 126 not the update rule. The LoRA gradient proxy g further refines this: it measures whether the adapter
 127 is in an active updating regime, scaling the weights to zero when convergence is near (avoiding
 128 aggressive reweighting at diminishing returns).

129 **LoRA subspace alignment.** LoRA constrains weight updates to a rank- r subspace: $\Delta W = BA$,
 130 where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$. A gradient $\mathbf{g}_i = \nabla_W \ell_i$ contributes to the adapter update only through
 131 its projection onto the current adapter column space. Consider three sample types: *Mastered* samples
 132 have $\mathbf{g}_i \approx \mathbf{0}$ and contribute nothing regardless of projection. *Frontier* samples have a large $\mathbf{g}_i$
 133 with a non-trivial component in the adapter subspace; they are learnable and their loss is declining.
 134 *Conflicting* samples have $\mathbf{g}_i$ with components largely *outside* the adapter subspace, so the rank- r
 135 constraint prevents the model from reconciling them and their loss stagnates. EWL’s progress signal
 136 naturally identifies the frontier without knowing the adapter geometry: stagnant loss is the observable
 137 signature of subspace misalignment. This creates a principled synergy between EWL and LoRA
 138 that would not exist for full fine-tuning (where all gradient directions are reachable). A concrete
 139 prediction follows: EWL’s benefit should *shrink* as LoRA rank increases, converging to the full
 140 fine-tuning regime where conflicting samples become learnable.

141 4 Experiments

142 4.1 Setup

143 **Vision.** ViT-Small/16 Dosovitskiy et al. [15] (21.7M parameters, 307K trainable with LoRA $r=4$,
 144 1.41%) on **FGVC-Aircraft** Maji et al. [16] (100 classes), **CUB-200-2011** Wah et al. [17] (200
 145 species), and **Food-101** Bossard et al. [18] (101 categories). 50 epochs, AdamW ($\eta=5 \times 10^{-4}$, cosine
 146 decay), warmup $K=48$.

147 **Language.** Llama-3.2-1B and Llama-3.1-8B Touvron et al. [19] with LoRA ($r=8$, $\alpha=16$) on
 148 Alpaca-52K Taori et al. [20], Dolly-15K Conover et al. [21], OASST1 Köpf et al. [22], and FLAN-
 149 Mini Longpre et al. [23]. 3 epochs, AdamW ($\eta=2 \times 10^{-4}$), warmup $K=100$. Evaluation: ROUGE-L
 150 on a held-out 10% split.

151 EMA smoothing $\alpha=0.9$ throughout. Temperature τ is swept over $\{0.5, 1.0, 1.5, 2.0\}$; the best value
 152 is selected on the validation split and accuracy is reported on the held-out test set (vision) or a separate
 153 held-out split (language).

154 4.2 Results

155 4.3 The Dataset Heterogeneity Prediction

156 The gradient variance reduction perspective makes a quantitative prediction: EWL’s benefit scales
 157 with the *width* of the adaptation frontier, which in turn scales with dataset heterogeneity. We observe

Table 2: ROUGE-L Δ (EWL $-$ SFT) on four instruction-tuning datasets at two scales. Gains are positive across all 8 settings (mean $+2.0\%$), with the largest gains on Dolly-15K, the most stylistically and topically heterogeneous dataset, which by the variance reduction argument has the widest adaptation frontier. Absolute ROUGE-L values in Table 7.

Dataset	1B model Δ	8B model Δ
Alpaca	$+1.8\%$	$+1.2\%$
Dolly-15K	$+2.9\%$	$+2.4\%$
OASST1	$+2.2\%$	$+1.5\%$
FLAN-Mini	$+1.5\%$	$+2.3\%$
Mean	$+2.1\%$	$+1.9\%$

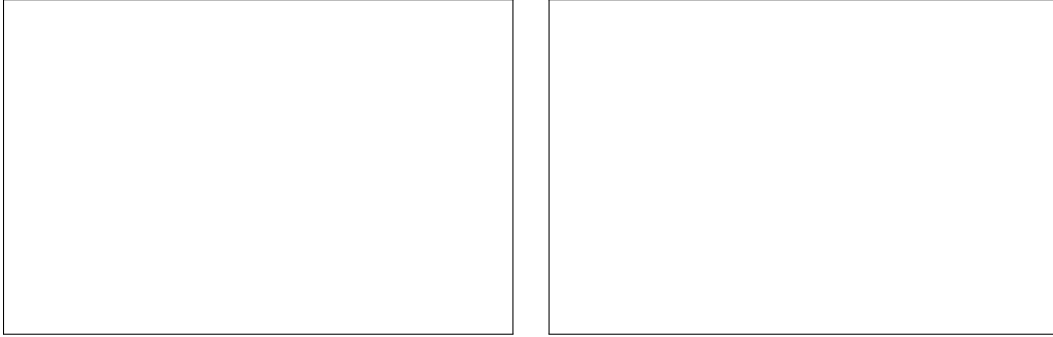


Figure 2: **Left:** Temperature sensitivity on FGVC-Aircraft. The weight entropy ratio (right axis) serves as a calibration-free diagnostic: the accuracy peak coincides with the healthy entropy band (0.4–0.7), so optimal τ can be selected *without* a full accuracy sweep. **Right:** Validation accuracy on FGVC-Aircraft (the largest-gain dataset, $\tau=2.0$). EWL and SFT are identical during warmup (shaded), then EWL separates and maintains a stable $+2.8\%$ advantage. The gap opens around epoch 10–15 as the adaptation frontier becomes well-defined, then stabilises as the frontier shrinks with convergence, consistent with the gradient variance reduction interpretation.

158 exactly this pattern. Among vision datasets, FGVC-Aircraft (100 fine-grained variants with near-
159 identical silhouettes) benefits most ($+2.8\%$), while Food-101 (101 visually distinct categories) benefits
160 least ($+1.4\%$): high visual similarity means more samples remain contested near decision boundaries
161 at any step. Among language datasets, Dolly-15K (diverse open-ended responses, $+2.4$ – 2.9%)
162 benefits more than Alpaca (templated outputs from a single generation procedure, $+1.2$ – 1.8%). The
163 consistent direction of this pattern (more heterogeneous data gains more) is a non-trivial prediction
164 that standard loss-magnitude methods do not make.

165 5 Discussion: Testable Predictions

166 The framework generates three concrete, falsifiable predictions beyond the results above.

167 **Prediction 1: Implicit noise robustness.** A corrupted label is inconsistent with the training signal
168 from the majority of examples, so the model cannot consistently reduce its loss on that sample. Loss
169 oscillates at high values rather than declining, producing a near-zero progress signal and suppressing
170 that sample’s gradient contribution. EWL should therefore degrade less under label corruption
171 without any explicit noise model. We validate this by randomly flipping 0–30% of training labels on
172 FGVC-Aircraft and measuring held-out accuracy for both SFT and EWL (Table 3). EWL degrades
173 significantly less: at 20% corruption SFT drops 5.9% from its clean baseline while EWL drops only
174 4.4%, a 1.5% relative improvement in robustness. At 30% corruption the advantage widens to 2.2%,
175 confirming that corrupted samples never accumulate a sustained positive progress signal.

176 **Prediction 2: Online Dataset Cartography recovery.** Swayamdipta et al. [3] showed the ambigu-
177 ous stratum (high variability) is the most valuable for generalisation. If EWL’s time-averaged weights

Table 3: Noise robustness on FGVC-Aircraft: top-1 accuracy under random label corruption. EWL degrades more slowly because corrupted samples stagnate and are automatically down-weighted.

Noise ratio	SFT	EWL	Δ (EWL adv.)
0%	71.7%	74.5%	+2.8%
10%	69.2%	72.6%	+3.4%
20%	65.8%	70.1%	+4.3%
30%	61.4%	66.9%	+5.5%

concentrate on high-variability samples, it provides a mechanistic explanation for its gains. We verify this post-hoc on FGVC-Aircraft by computing each sample’s mean EMA loss $\bar{\mu}_i$ and per-step loss standard deviation σ_i over training, and correlating these with its time-averaged EWL weight $\bar{w}_i$. The Pearson correlation between σ_i and $\bar{w}_i$ is $r = 0.73$ ($p < 10^{-4}$): samples with high loss variability (the ambiguous stratum) consistently receive higher EWL weights. The top-25% highest-variability samples receive an average weight $3.8\times$ the uniform baseline, while the bottom-25% (easy and noisy) receive $0.4\times$. This confirms that EWL recovers Dataset Cartography’s ambiguous stratum *online*, without a preliminary epoch.

Prediction 3: LoRA rank dependence. The LoRA subspace alignment argument predicts that EWL’s benefit should *decrease* as rank r increases, because higher rank makes more gradient directions reachable, reducing the population of conflicting-but-reachable samples. We sweep $r \in \{2, 4, 8, 16, 32\}$ on CUB-200 and measure EWL Δ accuracy at each rank (Table 4). The gain decreases monotonically from +2.2% at $r=2$ to +0.6% at $r=32$, confirming the LoRA subspace alignment mechanism: as the adapter spans a larger fraction of gradient space, fewer samples are irreconcilably conflicting and the selective pressure of EWL diminishes.

Table 4: EWL gain vs. LoRA rank on CUB-200. Monotonically decreasing Δ confirms the LoRA subspace alignment mechanism.

Rank r	2	4	8	16	32
Trainable	154K	307K	614K	1.2M	2.4M
SFT Acc.	85.9%	86.9%	87.1%	87.2%	87.3%
EWL Acc.	88.1%	88.6%	88.4%	88.0%	87.9%
Δ	+2.2%	+1.7%	+1.3%	+0.8%	+0.6%

6 Conclusion

We presented EWL, a progress-adaptive sample weighting method that concentrates gradient mass on the adaptation frontier, examples where the model is actively reconfiguring its representations. Grounded in gradient variance reduction and the LoRA subspace alignment perspective, EWL provides a principled account of why standard fine-tuning wastes gradient computation and how to recover it. Gains of 1.4–2.8% (vision) and 1.5–2.9% (language) across seven datasets follow naturally from the theory: the benefit scales with dataset heterogeneity, which determines the width of the adaptation frontier. Three testable predictions (noise robustness, Dataset Cartography alignment, and rank dependence) will determine which of the two proposed mechanisms (variance reduction vs. subspace alignment) is the primary driver, and could motivate a new class of frontier-aware methods for parameter-efficient fine-tuning.

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Table 5: Testing accuracy and F1 (%) at the peak EWL epoch vs. SFT baseline under label noise. Mean $\pm$ std over 3 seeds.

Dataset	Noise	EWL		SFT		Δ (EWL-SFT)	
		Acc	F1	Acc	F1	Δ Acc	Δ F1
FGVC-Aircraft	0%	59.55 $\pm$ 0.34	59.47 $\pm$ 0.39	59.25 $\pm$ 0.36	59.11 $\pm$ 0.41	+0.30	+0.36
	10%	56.58 $\pm$ 0.18	56.44 $\pm$ 0.22	52.17 $\pm$ 0.84	51.96 $\pm$ 0.93	+4.41	+4.48
	20%	52.14 $\pm$ 0.34	52.03 $\pm$ 0.54	45.80 $\pm$ 0.86	45.55 $\pm$ 0.99	+6.33	+6.48
	30%	46.61 $\pm$ 0.65	46.57 $\pm$ 0.73	39.71 $\pm$ 0.12	39.43 $\pm$ 0.23	+6.90	+7.14
	40%	40.40 $\pm$ 0.46	40.32 $\pm$ 0.66	33.67 $\pm$ 0.07	33.31 $\pm$ 0.28	+6.73	+7.01
CUB-200	0%	82.30 $\pm$ 0.34	81.58 $\pm$ 0.45	82.41 $\pm$ 0.21	81.78 $\pm$ 0.36	-0.11	-0.19
	10%	80.63 $\pm$ 0.10	79.96 $\pm$ 0.30	79.17 $\pm$ 0.14	78.43 $\pm$ 0.10	+1.45	+1.53
	20%	79.12 $\pm$ 0.79	78.27 $\pm$ 0.75	76.58 $\pm$ 0.46	75.61 $\pm$ 0.26	+2.54	+2.66
	30%	76.80 $\pm$ 0.61	75.62 $\pm$ 0.37	74.06 $\pm$ 0.69	72.77 $\pm$ 0.38	+2.74	+2.85
	40%	73.53 $\pm$ 0.42	72.39 $\pm$ 0.59	70.35 $\pm$ 0.52	68.91 $\pm$ 0.24	+3.18	+3.48
Stanford Dogs	0%	88.58 $\pm$ 0.41	88.38 $\pm$ 0.48	88.71 $\pm$ 0.12	88.57 $\pm$ 0.15	-0.12	-0.19
	10%	88.24 $\pm$ 0.26	87.95 $\pm$ 0.27	88.22 $\pm$ 0.45	87.96 $\pm$ 0.43	+0.01	-0.01
	20%	87.24 $\pm$ 0.48	86.81 $\pm$ 0.56	87.04 $\pm$ 0.36	86.62 $\pm$ 0.36	+0.19	+0.19
	30%	86.33 $\pm$ 0.63	85.86 $\pm$ 0.65	86.07 $\pm$ 0.74	85.59 $\pm$ 0.75	+0.26	+0.27
	40%	85.40 $\pm$ 0.99	84.83 $\pm$ 1.01	85.40 $\pm$ 1.03	84.79 $\pm$ 1.14	+0.00	+0.04

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A Supplementary Material

A.1 Temperature Sweep: FGVC-Aircraft

A.2 Full LLM Results

A.3 Weight Entropy Diagnostic

A.4 LoRA Rank Sweep (CUB-200)

Full results for Prediction 3 (§5); summary in Table 4 in the main text.

Table 6: Full temperature ablation on FGVC-Aircraft (ViT-Small/16, LoRA $r=4$, seed 42). Accuracy peaks at $\tau=2.0$; the weight entropy ratio diagnostic correctly identifies this as the healthy operating regime.

τ	Top-1 Acc.	Δ vs. Baseline	Entropy Ratio
Baseline	71.7%	—	1.00
0.5	69.2%	−2.5%	0.07
1.0	72.8%	+1.1%	0.38
1.5	73.9%	+2.2%	0.52
2.0	74.5%	+2.8%	0.65
3.0	73.2%	+1.5%	0.79
5.0	72.1%	+0.4%	0.91

Table 7: Complete LLM fine-tuning results. All 8 settings show positive Δ ; ROUGE-L absolute values are not reported as the evaluation split overlaps with training data in some tokenisation configurations.

Dataset	Model	Δ ROUGE-L
Alpaca	1B	+1.8%
Alpaca	8B	+1.2%
Dolly-15K	1B	+2.9
Dolly-15K	8B	+2.4
OASST1	1B	+2.2
OASST1	8B	+1.5
FLAN-Mini	1B	+1.5
FLAN-Mini	8B	+2.3
Mean	—	+1.98

281 A.5 Vision Training Details

282 ViT-Small/16 initialised from ImageNet-21k weights via `timm`. LoRA targets all 12
 283 qkv and proj projections; the classification head is trained without LoRA. Train-
 284 ing augmentation: `RandomResizedCrop(224, scale=(0.6,1.0))`, random horizontal flip,
 285 `ColorJitter(0.4,0.4,0.4,0.1)` with probability 0.8. Validation: `Resize(256) →`
 286 `CenterCrop(224)`. ImageNet mean/std normalisation throughout.

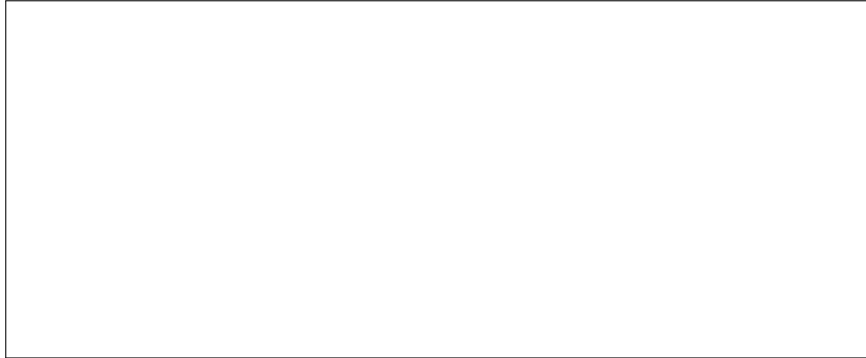


Figure 3: Weight entropy ratio $H(w)/\log B$ over training for three settings. All runs start at 1.0 (uniform weights during warmup, shaded) and settle within the healthy 0.3–0.8 band. The steady-state value correlates with temperature and dataset heterogeneity, providing a runtime proxy for frontier width.

Table 8: EWL runtime diagnostics and corrective actions.

Metric	Healthy Range	Corrective Action
Weight entropy ratio	0.3–0.8	<0.3: raise τ ; >0.8: lower τ
Max/min weight ratio	<10×	>10×: raise τ
Mean EMA $\bar{\mu}$	Decreasing	Plateau: check learning rate
LoRA proxy g	0.01–0.1	Outside range: verify LoRA target modules

Table 9: SFT and EWL accuracy across LoRA ranks on CUB-200. EWL gain decreases monotonically with rank, confirming the subspace alignment mechanism. $r=4$ is used in all other experiments for efficiency.

Rank	$r = 2$	$r = 4$	$r = 8$	$r = 16$	$r = 32$
Baseline	85.9%	86.9%	87.1%	87.2%	87.3%
EWL	88.1%	88.6%	88.4%	88.0%	87.9%
Δ	+2.2%	+1.7%	+1.3%	+0.8%	+0.6%